

# Juice Gets a Gravity Boost from Earth

From: ScienceDaily Most Popular Mined: 2026-10-03 3 min read

How can a **spacecraft** speed up without burning lots of fuel? It can get help from a planet's **gravity**. On September 28, the European Space Agency's Juice **spacecraft** used Earth for just such a boost on its journey toward Jupiter.

This move is called a **gravity** assist. A carefully chosen path past a planet lets its **gravity** change a **spacecraft**'s speed and direction. Earth's pull turned Juice's path by 20 degrees and increased its speed by 3.5 kilometers per second.

At its closest point, Juice passed 8,640 kilometers above the Indian Ocean. Its monitoring cameras took a series of pictures. During the four weeks before the encounter, controllers needed just one small adjustment to its **course**.

Saving fuel matters. Juice will need fuel later for observations around Jupiter and its icy moons. The **spacecraft** is scheduled to make 35 close passes by Ganymede, Callisto and Europa.

Earth also served as a practice target for Juice's science tools. Teams planned activities for its 10 **instruments**, including checks to make sure their measurements work as expected. Such checks are called **calibration**.

The team had to choose activities carefully. While Juice was in Earth's shadow, it relied only on battery power. The visit also gave scientists a chance to learn how the **instruments** behave in space.

For Juice, a trip past home was both a helpful push and a **rehearsal** for the science ahead.

## Word Treasure

**gravity** -- The pull that objects with mass have on one another.

**spacecraft** -- A vehicle designed to travel in space.

**course** -- The path a vehicle follows.

**instruments** -- Tools used to make scientific measurements.

**calibration** -- Checks and adjustments that help a tool measure correctly.

**rehearsal** -- Practice to prepare for a later activity.

## Background you need

A gravity assist uses a planet's gravity to change a spacecraft's speed and direction. Calibration helps teams check that scientific instruments measure as expected.

## Break it down

Opens with a question about gaining speed while using little fuel.  
Explains the gravity assist and describes Juice's Earth encounter.  
Connects saved fuel with planned observations of Jupiter's moons.  
Describes instrument checks and battery use in Earth's shadow.  
Ends by describing the encounter as preparation for future science.

## Why it matters

The Earth visit helped Juice save fuel and gave teams a chance to check the instruments that will study Jupiter's icy moons.

## Test what you remember

**1. Which planet supplied Juice's gravity boost on September 28?**

- ☐ (A) Mars
- ☐ (B) Earth
- ☐ (C) Saturn
- ☐ (D) Jupiter

**2. What did Earth's gravity change during the encounter?**

- ☐ (A) Juice's size and shape
- ☐ (B) Juice's speed and direction
- ☐ (C) Juice's batteries and fuel tanks
- ☐ (D) Juice's instruments and cameras

**3. Where was Juice at its closest point to Earth?**

- ☐ (A) Above the Atlantic Ocean
- ☐ (B) Above the Arctic Ocean
- ☐ (C) Above the Indian Ocean
- ☐ (D) Above the Pacific Ocean

**4. Why is saving fuel useful for Juice?**

- ☐ (A) It needs fuel to create new moons.
- ☐ (B) It needs fuel to strengthen Earth's gravity.
- ☐ (C) It needs fuel for later observations.
- ☐ (D) It needs fuel to enlarge its cameras.

**5. What are the instrument checks called?**

- ☐ (A) Navigation
- ☐ (B) Calibration
- ☐ (C) Communication
- ☐ (D) Transportation

**6. What supplied Juice's power while it was in Earth's shadow?**

- ☐ (A) Jupiter's moons
- ☐ (B) Its cameras
- ☐ (C) Earth's oceans
- ☐ (D) Its batteries

Answer key (try the quiz first!): 1: B 2: B 3: C 4: C 5: B 6: D

**MY THOUGHTS (at least 20 words):**